# How Machines Learn

Training, testing, and the honesty check — the engine inside the black box.

Wednesday 1:15–3:20 PM · How Machines Learn



**Two students revise for the same exam. Student A studies last year's paper again and again. Student B studies the ideas behind the questions. The final exam asks NEW questions. Who passes — and why?**

*We return to this question at the end of the session.*

# 3 Today's Session Clock

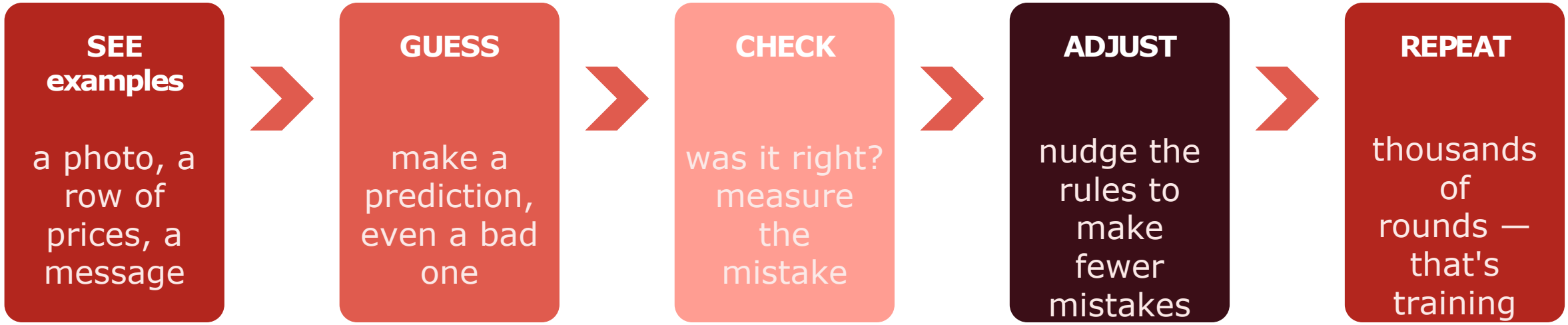
| Block | Time | What happens |
|---|---|---|
| Welcome & recap | 1:15–1:20 | The AI intoduction in 3 sentences; data = fuel |
| Lecture 1 — The learning loop | 1:20–1:55 | Training, features, labels — how a machine learns |
| Break | 1:55–2:10 | 5 minutes |
| Lecture 2 + Demo | 2:10–2:15 | Train vs test split; live: teach a mini model |
| Break | 2:15–2:20 | 5 minutes |
| Lab — Be the AI | 2:20–2:50 | Train a paper model: features → rule → test accuracy |
| Quiz + review | 2:50–3:10 | 5 questions, instant review |
| Summary & homework | 3:10–3:15 | Bring your dataset — it becomes the lesson |
| Optional extension | 3:15–3:20 | Second dataset card · stretch accuracy questions |

# 4 By the end of this session you can

*By the end of this session you can…*

- Say what training means: showing examples and correcting mistakes
- Pick out the features and the label in any dataset
- Explain why we test on data the model has never seen
- Compute accuracy: correct guesses ÷ total guesses
- Spot when high accuracy is a lie (memorising instead of learning)

# 4 The learning loop




*Training = loop. The machine gets better by checking its own mistakes, again and again.*

# 4 Training — learning from examples

**ANALOGY — think of it like…**

*A new shopkeeper learns prices: a customer asks for 'caano', the shopkeeper guesses, the customer corrects him. After a hundred corrections he just knows. Nobody gave him a price list for every possible sentence — he trained.*

**FORMAL DEFINITION**

Training is the process of showing a machine many labelled examples and repeatedly adjusting its internal rules until its guesses are mostly right.

**SOMALI CONTEXT — halka aan ka helno**

Your keyboard trained on the messages YOU type: every autocorrect you accept or reject is one training round.

# 4 Features vs label — what the machine sees

## FEATURES (what you measure)

- The inputs — what the machine looks at
- Tomato: colour, size, firmness
- Message: words, sender, time
- Photo: pixels, shapes, edges
- More useful features → better guesses

## LABEL (the answer)

- The output — what we want to predict
- Tomato: ripe / not ripe
- Message: spam / not spam
- Photo: cat / dog / car
- Labels come from humans or history

# 4 Accuracy — how good is the guess?

**ANALOGY — think of it like…**

*An exam: 10 questions, 7 correct → 70%. Simple. A model: 100 test predictions, 92 correct → 92% accuracy.*

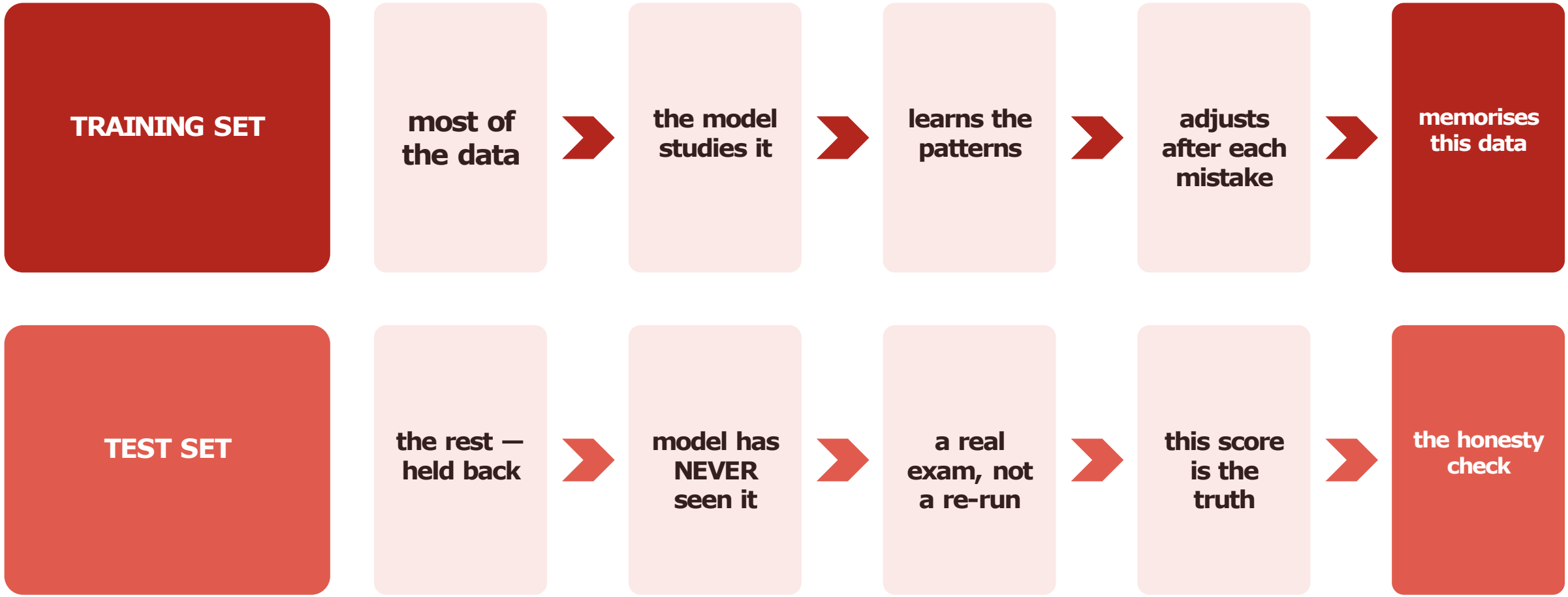
**FORMAL DEFINITION**

Accuracy = correct predictions ÷ total predictions, shown as a percentage. 92 of 100 = 92%.

**SOMALI CONTEXT — halka aan ka helno**

A weather app (illustrative): if it predicts rain for 10 days and 7 days actually rain, that's 70% accuracy — useful, not perfect.

# 4 The honesty check: train / test split




*Never test on the same data you trained on. That's like marking your own homework.*

# 4 When accuracy lies

| Situation | Training accuracy | Test accuracy | What's happening |
|---|---|---|---|
| Student A: memorised the old paper | 100% | 40% | Memorised, didn't learn — fails new questions |
| Student B: learned the ideas | 85% | 84% | Learned patterns — does well on new questions |
| Model trained on 80 rows + tested on the same 80 | 99% | 99% | The test is fake — it already saw the answers |
| Honest model: 80 train / 20 held back | 88% | 86% | Both scores are real; small drop = healthy |

*The gap between training and test accuracy tells you if the model learned — or memorised.*

# 4 Live — teach the assistant a rule

**YOU TYPE**

```
Prompt: 'Here are 3 examples of Somali proverbs translated to English: (1) … (2) … (3) … . Now translate this new proverb the same style: …'
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**AI RETURNS**

(Live output — the assistant follows your 3 examples as the 'training set' and translates the new one in the same style.)

**Takeaway: Even a few examples can 'train' an assistant for a task — this is few-shot prompting, the same loop: examples → pattern → apply to new.**

ACTIVE LEARNING PAUSE

# Your keyboard suggests the next word, you accept or reject it. (a) What are its features? (b) What is its label? (c) Which data is its test set?

**Think · Pair · Share**

3 minutes — buddy up, compare answers

# 4 Four learning mistakes to avoid

| The mistake | Why it happens / the fix |
| --- | --- |
| 'More data always means better AI' | More RAM doesn't make a better student. Quality + clean + relevant data beats volume (W03 returns). |
| '100% accuracy on the training set = perfect model' | That's memorisation, not learning. The TEST set score is the truth. |
| 'AI learns instantly, like magic' | Training = millions of small adjustments. The magic is thousands of corrected mistakes. |
| 'Testing on the same data is fine' | No — that's marking your own homework. Held-back data is the only honest exam. |

# 4 Lab — Be the AI

**OBJECTIVE**

Train a tiny paper model on tomato data: pick features, build a rule from the training set, then test it on unseen tomatoes and compute its accuracy.

**STEPS**

- 1. Get the dataset card (12 tomatoes: colour, firmness, size + ripe/not-ripe)
- 2. Use ONLY the first 9 as training — build your rule ('if colour is X and firmness is Y → ripe')
- 3. Lock the rule away. Then open the 3 test tomatoes — predict with your rule
- 4. Reveal the answers. Compute accuracy: correct ÷ 3
- 5. Repeat with the whole class's rules — compare accuracy scores

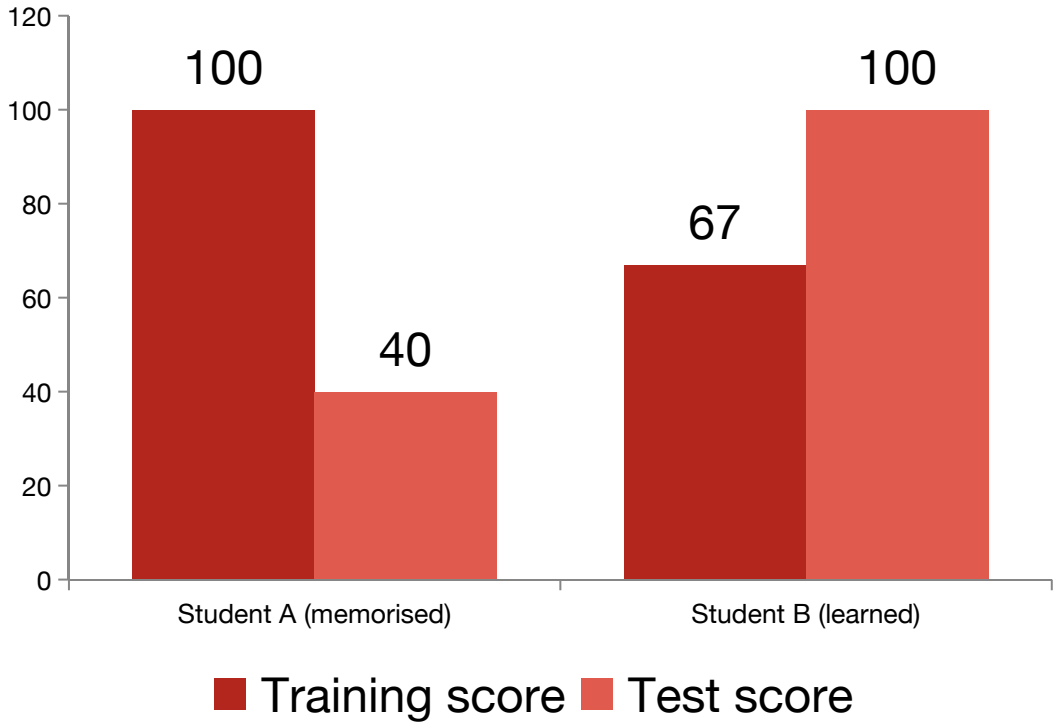
**DELIVERABLE — Your rule written as one sentence + the test predictions and accuracy ( /3 correct)**

# 4 Lab checkpoint

**STOP — answer before continuing**

**Your rule scored 100% on the 9 training tomatoes. Your classmate's rule scored 67% on training but 100% on the 3 test tomatoes. Whose rule would you deploy on a farm — and why?**

*Tell your buddy. Then show the instructor.*



# Independent challenge

**SOLO · NO HELP · 10 MIN**

- Take ONE new tomato from the extra card — no peeking at its answer
- Predict ripe / not-ripe with YOUR rule, then reveal
- Write one sentence: is 3 test examples enough to trust your rule?
- Make a 4th test example of your own (invent the features) and swap with your buddy

*Done? Show the instructor. Explain it aloud — that is the check.*

# 4 Quick Quiz — Week 4

*5 questions · 15 minutes · answers stay with the instructor*

Q1 — Define training in one sentence (your own words).

Q2 — A dataset of tomatoes lists colour, size and ripeness. Which are features and which is the label?

Q3 — Why must the test set contain data the model has never seen?

Q4 — A model is 95% accurate on its training data and 55% on its test data. What does the gap tell you?

Q5 — 10 test predictions, 8 correct — what is the accuracy?

# 4 Your project

**THIS WEEK FEEDS YOUR GROUP PROJECT**

- The train/test split you used on paper today is the SAME method your team uses in Week 12
- Your project's test log: prototype → test on unseen cases → accuracy → fix
- Start collecting your dataset with a test set in mind: keep ~20% for the final check

Project pipeline: choose a problem (W11) → build (W11–W12) → test (W12) → document (W13) → defend (W14).

# 4 Today's takeaways

- Training = loop: see → guess → check → adjust → repeat
- Features are what you measure; the label is the answer
- Accuracy = correct ÷ total — but only on the TEST set
- Never test on training data — that's marking your own homework
- A big train/test gap = memorising, not learning



# Own it.

*Learn like a student who studies ideas, not answers.*

**Next week: AI assistants & prompting — how to steer the tool you trained with examples.**